

Market structure in European electricity and gas, 2013–2023: Germany in comparative perspective

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23 September 2026

Abstract

This note describes electricity-market concentration and gas-retailer density for countries that report in Eurostat’s energy market indicators, from 2013 to 2023, with Germany as the reference country. Concentration is the percentage of national electricity capacity, and of national electricity generation, held by companies that each have at least 5 percent of that market. Gas-retailer density is the number of gas retailers per terawatt-hour of final natural-gas consumption. Germany’s cumulative capacity share fell from 67.0 percent to 52.5 percent, and its cumulative generation share fell from 74.0 percent to 57.1 percent. Cross-country medians fell as well, and Germany stayed near the middle of both rankings. In 2023 Germany had the largest raw count of gas retailers, 973, and ranked 11th of 30 on retailers per terawatt-hour. The comparison is descriptive. It does not identify the effect of any policy.

1 Introduction

The question is where Germany sat, between 2013 and 2023, in the European distribution of two electricity-market concentration measures, and in the distribution of gas retailers once the count is scaled by the size of the gas market.

Germany is the reference country because that is the comparison this project set out to make, and because an unscaled ranking of gas retailers puts Germany first for a mechanical reason: larger markets have more retailers. The paper reports Germany against the annual cross-country median and Germany’s rank. Rank 1 is the highest value. Ties take the minimum rank. The median and the interquartile range include Germany.

Two readings are fixed before the ranks. A concentration share “fell” or “rose” only if the change from 2013 to 2023 exceeds 2 percentage points. Retailer density “fell” or “rose” only if the 2023 level is at least 10 percent away from the 2013 level.

The note does not estimate a policy effect. It does not measure prices, grid quality, or social cohesion. Electricity generation concentration and the number of gas retailers are different markets. An exploratory regression of one on the other is reported in Appendix A and is not part of the abstract, because it does not survive the rule set out there.

2 Data and measurement

Market structure comes from Eurostat’s energy market indicators, table `nrg_ind_market` (Eurostat, 2026a). The collection is voluntary. Some cells are unavailable or confidential, and the published mark for both is the same (Eurostat, 2025). In this extract that mark is stored as missing, not as zero. A reported zero stays zero. The window is 2013–2023. The 2024 column is almost empty and is dropped. There is no EU aggregate in the file. The reporters are EU members together with other countries in the extract, including Turkey, Georgia, Moldova, and several Western Balkan countries. Norway does not report the headline series. Greece is `EL`. Georgia is `GE` and is not part of Germany.

Table 1 lists the series. Official labels are from the Eurostat vocabulary `indic_nrgm` (European Environment Agency, 2021). That vocabulary has no prose definitions. The percent-versus-count reading is fixed from the unit and from one country-year. In Germany in 2013, the capacity series is 67 percent and the largest firm’s capacity share is 29 percent. The generation series is 74 percent and the largest firm’s generation share is 32 percent. The gas series is 825. The group share is larger than the largest firm on both electricity margins, which is what a cumulative share does. The gas figure is a count of retailers. These series are not prices, and the retailer count is not a measure of grid quality.

Table 1: Series used in the comparison

Code	Unit	Product	Official label and role
<code>CMPY_ECAP5</code>	Percent	Electricity	Companies with at least 5% of electricity capacity. Cumulative capacity share.
<code>CMPY_EG5</code>	Percent	Electricity	Companies with at least 5% of electricity generation. Cumulative generation share. Not a gas series.
<code>LCMPY_IECAP</code>	Percent	Electricity	Largest company, installed electricity capacity.
<code>LCMPY_EG</code>	Percent	Electricity	Largest company, electricity generation.
<code>GRTL</code>	Number	Natural gas	Gas retailers. Used as an input to density.

Eurostat’s June 2025 *Statistics Explained* article reports the 2023 market share of the largest electricity generator as 85 percent in Cyprus, 78 percent in Croatia, and 74 percent in France (Eurostat, 2025). In this extract the largest-firm generation shares for those countries are 84.9, 78.4, and 73.9 percent. The series used here is that market share.

Final natural-gas consumption is the denominator for retailer density. Eurostat’s simplified energy balance, `nrg_bal_s`, has no single code for final consumption (Eurostat, 2026c). The denominator is the sum of final consumption for energy use (`FC_E`) and final consumption for

non-energy use (FC_NE), in gigawatt-hours, for natural gas. If either component is missing, the sum is missing. Retailers per terawatt-hour equal the retailer count divided by final consumption in terawatt-hours. Country-years below 1 TWh are out of the density ranking. That threshold was fixed before any rank was computed. The excluded rows are Cyprus, Montenegro, North Macedonia, and Kosovo in most years, all below 1 TWh, and Bosnia and Herzegovina in 2013, where consumption is missing. Forty-four country-years with a retailer count are excluded: 43 below the floor and one with missing consumption.

Population on 1 January, total, is a robustness denominator only (Eurostat, 2026b). Retailers per million inhabitants are computed on the same country-years that pass the 1 TWh screen. They are not the headline rank.

Each electricity chart uses every non-missing country-year of that series. Germany reports every year 2013–2023 on all five study series. Austria, the Netherlands, and Turkey never report the largest-firm shares, so they are absent from those ranks. The 2013–2023 window contains 385 country-years. Cumulative capacity share is non-missing in 382 of them, cumulative generation share in 376, and the retailer count in 371. The density sample has 327 country-years.

3 Electricity concentration

Germany’s cumulative shares fell, and the cross-country medians fell with them. Table 2 reports both endpoints. Figure 1 and Figure 2 plot Germany against the median and the interquartile range.

Series	Germany		Median		Germany’s rank	
	2013	2023	2013	2023	2013	2023
Capacity, firms with at least 5%	67.0	52.5	67.9	58.5	19 of 33	19 of 35
Generation, firms with at least 5%	74.0	57.1	76.1	63.4	20 of 33	21 of 35
Largest firm, capacity	29.0	18.8	52.7	36.5	25 of 31	28 of 32
Largest firm, generation	32.0	22.5	57.1	41.6	24 of 31	26 of 31

Shares are percentages. Rank 1 is the highest share. The median includes Germany. Source: Eurostat `nrg_ind_market`.

The cumulative capacity share fell by 14.5 percentage points, from 67.0 percent to 52.5 percent. The median fell from 67.9 percent to 58.5 percent. Germany was 0.9 percentage points below the median in 2013 and 6.0 below it in 2023. The rank stayed 19. Under the 2 percentage-point rule, the German share fell. The rank did not. Germany moved from the center of the distribution to a little below the center, while many countries remained much more concentrated: the 2023 upper quartile is still 78 percent.

The cumulative generation share fell by 16.9 percentage points, from 74.0 percent to 57.1 percent. The median fell from 76.1 percent to 63.4 percent. Germany was 2.1 percentage points below the median in 2013 and 6.3 below it in 2023. The rank moved from 20 of 33 to 21 of 35.

The largest-firm shares fell as well, and they sit far below the median throughout. Germany’s

largest firm held 18.8 percent of capacity in 2023, against a median of 36.5 percent (rank 28 of 32), and 22.5 percent of generation, against a median of 41.6 percent (rank 26 of 31). Of Germany's 52.5 percent cumulative capacity share in 2023, 18.8 percentage points are the largest firm. The rest is other firms that each hold at least 5 percent. Germany's market is less top-heavy than the median country's market. The cumulative share, which is close to the median, does not say that one firm dominates generation or capacity.

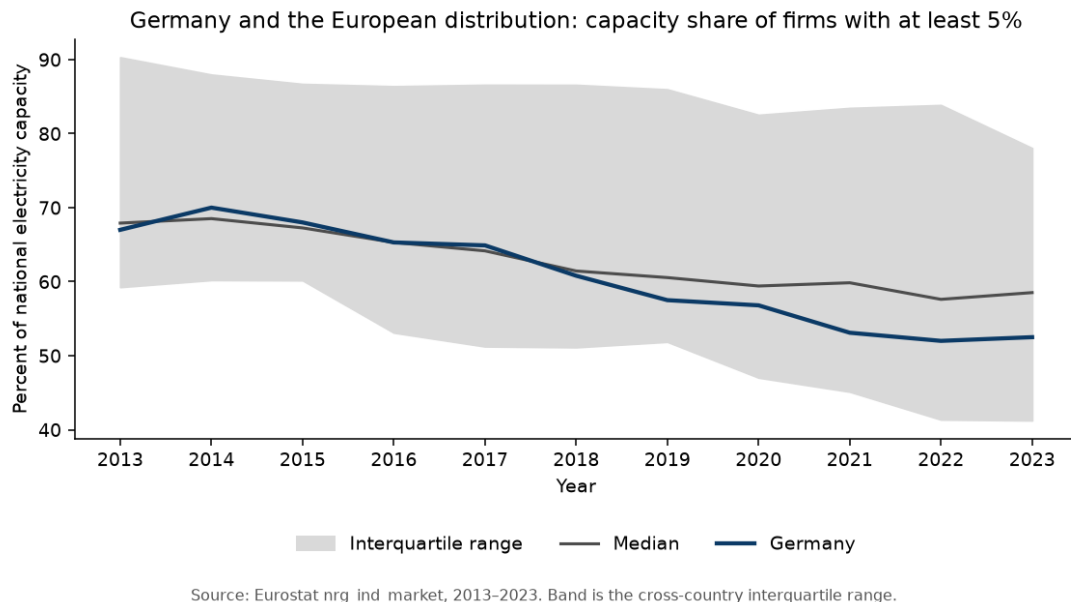


Figure 1: Cumulative capacity share of firms with at least 5 percent of national electricity capacity. The band is the cross-country interquartile range, including Germany. Source: Eurostat nrg_ind_market, 2013–2023.

4 Gas-retailer density

The raw count and the scaled count do not place Germany in the same part of the distribution. The count rose from 825 retailers in 2013 to 973 in 2023. Germany ranked 1 of 33 countries with a reported count in both years. That rank is market size.

Per terawatt-hour of final consumption, Germany rose from 1.28 retailers in 2013 to 1.76 in 2023, a rise under the 10 percent rule. The median rose from 0.77 to 1.15. Germany's rank moved from 8 of 28 to 11 of 30. Germany remained above the median and was not at the top. Figure 3 shows the path. Table 3 places several large markets in the 2023 cross-section. France, the Netherlands, and Turkey rank high on the raw count and low on density, because their consumption is large relative to the number of retailers.

The top of the 2023 density ranking is small markets just above the floor: Estonia (21 retailers, 1.9 TWh), Moldova (24 retailers, 3.7 TWh), and Bosnia and Herzegovina (7 retailers, 1.6 TWh). Density removes the advantage a large market has in a raw count. It is noisy when the market is only a few terawatt-hours. Germany's rank of 11 is the comparison this note uses. It is not a claim that those smaller markets have a deeper retail sector in absolute terms.

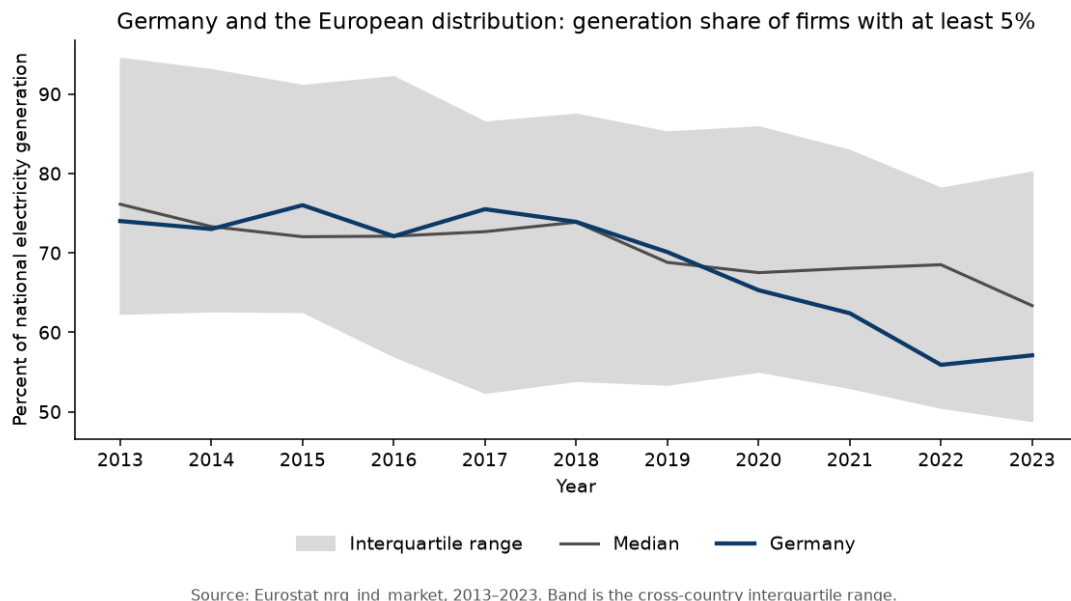


Figure 2: Cumulative generation share of firms with at least 5 percent of national electricity generation. The band is the cross-country interquartile range, including Germany. Source: Eurostat `nrg_ind_market`, 2013–2023.

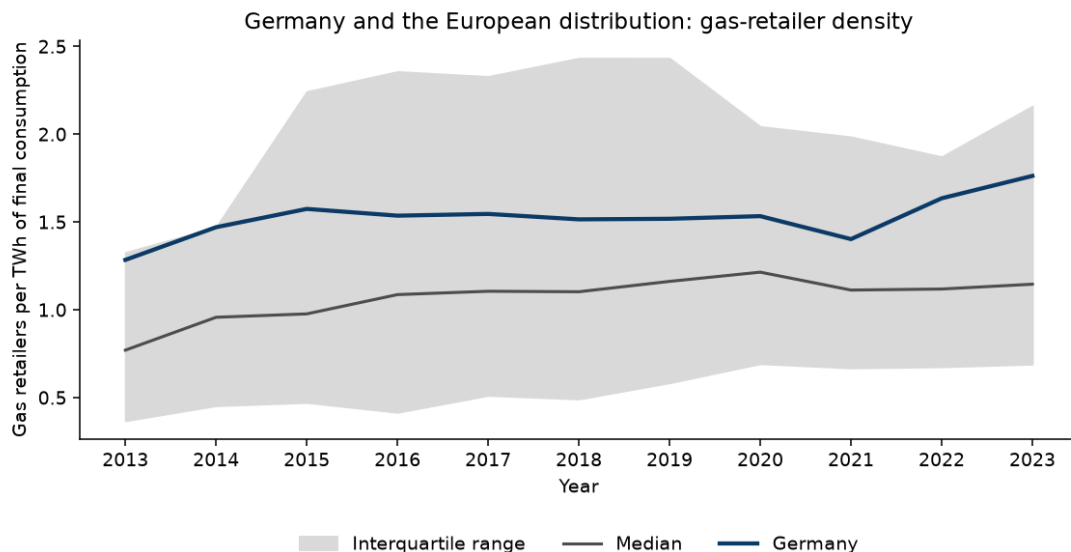
Scaled by population instead of consumption, Germany had 10.25 retailers per million inhabitants in 2013 (rank 2 of 28) and 11.71 in 2023 (rank 2 of 29). Per person, Germany stays near the top. Per terawatt-hour, it does not. The headline rank is the per-terawatt-hour rank, because the contract specified final consumption as the size measure. The population ranking shows that the choice of denominator changes the description.

5 What the comparison shows

On electricity, Germany deconcentrated over the decade, and so did the middle of the reporting sample. By 2023 the cumulative shares were about 6 percentage points below the median, with an unchanged capacity rank and a generation rank that moved by one place. The distinctive German fact on this margin is the largest firm, not the cumulative share. The largest firm’s capacity share, 18.8 percent, is well below the median of 36.5 percent.

On gas, Germany is first in the number of retailers and eleventh in retailers per terawatt-hour. Using the raw count as a ranking of market structure confuses the size of the market with the number of firms per unit of gas sold.

Nothing in this design says that a reform caused the decline in concentration or the rise in retailer density. The series are not linked to a dated policy change, and countries are not assigned to treatment and comparison groups. Appendix A reports the within-country association between generation concentration and retailer density. The association is about zero, and it stays out of the abstract.



Source: Eurostat `nrg_ind_market` and `nrg_bal_s` (FC_E + FC_NE). Countries with final gas consumption of at least 1 TWh.

Figure 3: Gas retailers per terawatt-hour of final natural-gas consumption. Country-years below 1 TWh are excluded. The band is the cross-country interquartile range, including Germany. Source: Eurostat `nrg_ind_market` and `nrg_bal_s`.

References

- European Environment Agency (2021). Vocabulary: Energy market indicator (`indic_nrgm`). Released 17 May 2021. Accessed 23 September 2026.
- Eurostat (2025). Electricity market indicators. Statistics Explained. Data extracted in June 2025. Accessed 23 September 2026.
- Eurostat (2026a). Energy market indicators (`nrg_ind_market`). Accessed 23 September 2026.
- Eurostat (2026b). Population on 1 January by age and sex (`demo_pjan`). Dataset updated 22 September 2026. Accessed 23 September 2026.
- Eurostat (2026c). Simplified energy balances (`nrg_bal_s`). Dataset updated 2 June 2026. Accessed 23 September 2026.

A Exploratory association

The pre-set rule was that a regression of retailer density on the cumulative generation share would enter the abstract only if the coefficient kept its sign and stayed within 50 percent of the baseline in every leave-one-out. The baseline is a country-year regression with country and year fixed effects. Standard errors are clustered by country. The sample is the density sample with a non-missing generation share: 326 country-years, 30 countries, and 11 years.

A one-percentage-point higher cumulative generation share is associated with 0.003 additional retailers per terawatt-hour. The cluster-robust 95 percent interval runs from -0.017 to

Table 3: Selected countries, gas retailers in 2023

Country	Retailers	Final use (TWh)	Raw-count rank	Density rank
Germany	973	552	1 of 33	11 of 30
Italy	634	348	2	10
Spain	130	155	3	22
Turkey	111	329	4	26
Poland	109	125	6	21
France	63	271	7	30
Netherlands	53	166	10	27

Density rank is retailers per terawatt-hour among country-years with final consumption of at least 1 TWh. Raw-count rank uses every country with a 2023 retailer count.

0.022. The within R^2 is 0.001. Leave-one-out coefficients run from -0.005 to 0.008 . Twenty-nine of 30 keep the baseline sign, and 25 of 30 stay within 50 percent of the baseline. The rule fails. The same sample with the unscaled retailer count as the outcome has a coefficient of -0.025 and an interval from -0.484 to 0.433 . Country fixed effects already remove permanent differences in market size, so that check is not where the size confound appears. The 2023 ranks in Table 3 are.

B Reproducibility

The panel, tables, and figures are produced by `scripts/run_pipeline.py`. Variable definitions are in `CODEBOOK.md`. The market extract is the repository file. Final consumption and population are downloaded from the Eurostat API into `data/external/` and cached.